APPROVED OPERATIONAL RUNBOOK: FXT-RUN-002

FXT Settlement Database Connection Pool Exhaustion & Thread Hung Runbook

Application: Settlement & Clearing (FXT-Settle)	**Service:** Oracle RAC / HikariCP Pool	**Region:** EMEA (London Core)
Environment: PROD	**Category:** Database & Transaction Processing	**Version:** 2.9.0
Owner / Team: FX Core Operations & APS Database Pod	**Last Updated:** 2026-09-02	**Doc ID:** FXT-RUN-002

1. Operational Symptoms & Issue Summary

Issue Summary: Application threads hanging on HikariCP connection checkout with ORA-00020 or connection timeout exceptions, freezing trade ticket processing.

Detectable Symptoms & Alerts:

- Log Exception: 'ConnectionTimeout: Connection is not available, request timed out after 30000ms'
- Log Error: 'ORA-00020: maximum number of processes (1500) exceeded'
- Alert: 'DB_POOL_EXHAUSTED: FXT_SETTLE_PROD pool 100% utilized (300/300 active connections)'
- Business impact: Post-trade booking queue rising over 5,000 unconfirmed transactions

2. Root Causes & Technical Diagnosis

- Uncommitted distributed transaction in batch settlement service holding exclusive row locks on T_TRADE_SETTLE.
- HikariCP connection leak due to missing try-with-resources in recent deployment of fx-settle-worker.
- Long-running ad-hoc reporting query spawned by Finance analyst blocking settlement redo logs.

3. Mandatory Safety Directives (DO NOT DO)

CRITICAL OPERATIONAL WARNING: **DO NOT perform an immediate hard kill (kill -9 or ALTER SYSTEM KILL SESSION IMMEDIATE) without recording the blocking SQL_ID and SID for root-cause forensic analysis.**
CRITICAL OPERATIONAL WARNING: **DO NOT restart Oracle RAC node 1 while node 2 is undergoing scheduled archive log sync.**
CRITICAL OPERATIONAL WARNING: **NEVER flush settlement buffer cache while intraday clearing window is open (14:00-16:30 London time).**

4. Step-by-Step Resolution Procedures

Step 1: Identify Blocking Sessions and Locked Objects

Run approved diagnostic query to extract blocking SID, serial, program, and wait event.

Command (SQL):

```
SELECT s.sid, s.serial#, s.username, s.status, s.program, s.blocking_session, s.event FROM v$session s WHERE s.blocking_session IS NOT NULL OR s.sid IN (SELECT blocking_session FROM v$session);
```

Step 2: Analyze Active Long-Running Statements

Check SQL text of the root blocker holding row exclusive locks.

Command (SQL):

```
SELECT sql_id, elapsed_time/1000000 elapsed_sec, cpu_time/1000000 cpu_sec, executions, sql_text FROM v$sql WHERE sql_id = '&BLOCKING;_SQL_ID';
```

Step 3: Terminate Orphaned / Hung Blocker Session

Execute approved safe session kill for the blocking SID identified.

Command (SQL):

```
ALTER SYSTEM KILL SESSION '&SID;,&SERIAL;' IMMEDIATE;
```

Step 4: Restart Worker Pods to Flush Stale Pool Leaks

Restart the settlement worker deployment gracefully one replica at a time.

Command (PowerShell):

```
kubectl rollout restart deployment/fxt-settle-worker -n fxt-prod
```

5. Post-Recovery Verification Procedures

V-1: Check HikariCP metrics: Active connections must drop below 40% (max 120/300).

V-2: Verify unconfirmed transaction queue on RabbitMQ/Kafka drops to < 50 items within 5 minutes.

V-3: Run SELECT COUNT(*) FROM v$session WHERE status = 'ACTIVE' AND username = 'FXT_APP';

6. Dependency-Aware Service Recovery Sequence

Ordered Sequence:

1. Terminate Blocking Session in Oracle RAC
   ➔ 2. Verify Database Lock Table Clear (v$lock count = 0 on T_TRADE_SETTLE)
   ➔ 3. Restart HikariCP Worker Deployment (fxt-settle-worker)
   ➔ 4. Resume Ingestion Consumer Queue

Explicit Stop/Escalate Gate: If ORA-00020 persists after terminating blocker, STOP immediately. Do NOT run mass session kill. Escalate to Senior Oracle DBA On-Call.

7. Edge Cases & Escalation Protocol

- **Edge Case:** 2-Phase Commit In-Doubt Transaction (ORA-01591): Requires DBA command ROLLBACK FORCE 'tx_id'.
- **Edge Case:** RAC Inter-Node Cache Fusion Contention (gc cr request): Verify interconnect bandwidth.

Primary Contact: Lead Production DBA (oracle-dba-oncall@fxtrading.internal | Phone: ext. 4402)	**Secondary / Bridge:** Settlement Operations Team (fxt-settle-ops@fxtrading.internal)
Severity / SLA: P1 Critical: 10-minute acknowledgement, 20-minute mitigation	**Escalation Condition:** Pool exhaustion unresolved after 10 minutes or queue exceeds 10,000 trades.